

Phishing Email Detection Model

A machine learning model built with Scikit-learn that classifies emails as **Phishing** or **Safe**.

Features

- Trains on a dataset of phishing and legitimate emails
- Extracts and analyzes email features:
 - Text features using TF-IDF (unigrams + bigrams)
 - URL features: number of links, IP-based links, suspicious domains (.xyz, .top, .click...), shortened links
 - Keyword and style features: urgent words (verify, password, suspend...), exclamation marks, uppercase ratio
- Classifies emails as "Phishing" or "Safe" using Logistic Regression
- Displays accuracy, confusion matrix and classification report
- Interactive mode to test your own email

Requirements

Python 3.8+

```
pip install scikit-learn pandas numpy scipy
```

How to run

```
python phishing_detector.py
# built-in sample dataset
python phishing_detector.py --csv
emails.csv    # your own dataset (columns:
text,label)
python phishing_detector.py --no-input
# skip the interactive prompt
```

Sample output

- Accuracy: 90% on the built-in sample dataset (32 emails, 70/30 train-test split)
- See `output.txt` for the full confusion matrix and classification report.

Note

The built-in dataset is a small sample for demonstration. The model can be trained on a larger

public dataset (e.g. Kaggle phishing email datasets)
by passing a CSV with `--csv`.

Expected outcome

The model classifies emails as Phishing or Safe
based on textual content and URL features.